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Productivity monitoring in building construction projects: a systematic review

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Received 7 March 2021
Revised 19 May 2021
Accepted 19 May 2021

Abstract

Purpose – The unique nature of the construction sector makes it fall behind other sectors in terms of productivity. Monitoring construction productivity is crucial for the construction project's success. Current practices for construction productivity monitoring are time-consuming, manned and error prone. Although previous studies have been implemented toward reducing these limitations, a gap still exists in the automated monitoring of construction productivity.

Design/methodology/approach – This study aims to investigate and assess the different techniques used for monitoring productivity in building construction projects. Therefore, a mixed review methodology (bibliometric analysis and systematic review) was adopted. All the related publications were collected from different databases, which were further screened to get the most relevant based on the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) criteria.

Findings – A detailed review was performed, and it was found that traditional methods, computer vision-based and photogrammetry are the most adopted data acquisition for productivity monitoring of building projects, respectively. Machine learning algorithms (ANN, SVM) and BIM were integrated with monitoring tools and technologies to enhance the automated monitoring performance in construction productivity. Also, it was observed that current studies did not cover all the complex construction job sites and they were applied based on a small sample of construction workers and machines separately.

Originality/value – This review paper contributes to the literature on construction management by providing insight into different productivity monitoring techniques.

Keywords Productivity monitoring, Construction, Productivity measuring, Monitoring techniques

Paper type Literature review

Data Availability Statement: All data, models, and code generated or used during the study appear in the submitted article.

The authors would like to thank Universiti Teknologi PETRONAS (UTP) for the support provided for this research.

Engineering, Construction and Architectural Management

0969-9988
DOI 10.1108/ECAM-03-2021-0211

1. Introduction

The construction industry plays an important role in supporting the economic development of any developed or developing country (Sabet and Chong, 2020; Van Tam *et al.*, 2021). It contributes to the economy by about 8%–10% on average in different countries (Dixit *et al.*, 2019). Construction job sites are dynamic environments that implicate various workers, equipment and machines operating simultaneously which make managing such job sites a challenging task (Zhang *et al.*, 2017). Monitoring, tracking activities and assuring that the whole project meets predicted production rates are the most fundamental duties of a manager (Sherafat *et al.*, 2020). The special nature of construction sites makes the use of standard industrial monitoring systems unpractical for the construction industry (Navon and Sacks, 2007). The absence of real-time information hindering managers' capability to monitor schedule, cost and other performance indicators, which reduces their ability to reveal or manage the variability and uncertainty inherent in project activities (Zhang *et al.*, 2009).

Despite the several challenges being faced by the construction industry, productivity is one of the most significant problems that cause cost and time overrun (Mahamid, 2013; Princy and Shanmugapriya, 2017). Productivity plays a significant role in evaluating the success of construction projects which reflects its impact on the construction industry (Mahamid, 2013). Also, productivity improvement plays a crucial role in lowering the costs of construction and increasing the revenue of the firms (Dixit *et al.*, 2019; Van Tam *et al.*, 2021). Construction firms can improve their performance, sustainability and competitive advantages in the market by adopting the best standard productivity practices (Marzuki *et al.*, 2012; Sabet and Chong, 2018). Therefore, productivity issues are described as accepted expanding considerations inside the construction sector (Dai *et al.*, 2009) and have been an interesting topic of research for the past five decades and that because the productivity growth in construction has lagged behind other sectors (Green, 2016).

Productivity has various definitions provided by several researchers; generally, it is defined as the ratio of output to input (Ayele and Fayek, 2019; Dixit *et al.*, 2017; Durdyev and Mbachu, 2018). Productivity rates are utilized by managers during scheduling to identify project life and decrease the cost of the resources (Alinaitwe Mwanaki, 2006; Palikhe *et al.*, 2019). In construction projects, productivity improvement will automatically improve its income, considering that 30%–50% of the total cost of a construction project is being spent on resources (Gunduz and Abu-Hijleh, 2020). Most of the factors that affect construction productivity can be improved by monitoring productivity effectively (Konstantinou and Brilakis, 2016). The monitoring methods can be divided into two major methods: traditional methods and automated methods. The traditional method depends on manual observations and statistical methods for collecting and analyzing data of different constructions operations and resources (Kim and Kim, 2018; Vahdatikhaki *et al.*, 2015). While automated methods have used new technologies and artificial intelligence for monitoring construction operations (Sherafat *et al.*, 2020). The current practices of traditional methods for collecting and monitoring data have some limitations for construction site monitoring, such as expensive, inaccurate, inefficient and time-consuming, which results in a slow flow of information and an inefficient decision making (Luo *et al.*, 2018; Naoum, 2015; Navon and Sacks, 2007; Omar *et al.*, 2018; Yang *et al.*, 2015).

While automated data collection for project performance control enables construction activities to be controlled and managed effectively and in real-time (Navon and Sacks, 2007). Several techniques like photogrammetry, vision-based, audio-based, tagged-studies and laser scanning are performed for automated monitoring of different construction activities, workers and machines (Mahami *et al.*, 2019; Sherafat *et al.*, 2020). Efficient productivity monitoring helps in assessing a project's performance and in enabling the identification of opportunities for improvement (Pradhan *et al.*, 2011). The selection of practical productivity monitoring tools is extremely valuable, which assures the reliability of data collection and reduces error probability (Langroodi *et al.*, 2021; Son *et al.*, 2015; Yang *et al.*, 2015). Ineffective monitoring of construction operations may lead to project failure, whereas effective monitoring enables timely corrective actions for delayed operations (El-Omari and Moselhi, 2011; Kopsida *et al.*, 2015).

Despite the great efforts by researchers in the field of construction productivity monitoring to improve the productivity of construction, the construction productivity is still low and does not improve within the past decades (Green, 2016) compared to the productivity of the industrial sector, which doubled in the same time (Changali *et al.*, 2015). Also, it is not improving over time with an annual 0.32% declining rate during the period from 2003 to 2012 (Teicholz, 2013). Furthermore, the construction industry does poorly completing megaprojects on time and budget; it is estimated that 98 % of megaprojects suffer cost overruns of more than 30% and 77% of these projects late at least by 40%; poor productivity is one of the most reasons for this poor records (Changali *et al.*, 2015). Choosing the appropriate techniques and tools for monitoring will improve construction productivity (Langroodi *et al.*, 2021; Pradhan *et al.*, 2011; Son *et al.*, 2015; Yang *et al.*, 2015). Therefore, it is very necessary to know the most appropriate techniques and tools for productivity monitoring in building construction projects.

Thus, this study aims to assess and review the knowledge domains of the available studies for the application of productivity monitoring (tools and techniques) and their implementation in construction building projects by adopting a mixed review methodology (bibliometric analysis and systematic review). Providing insight into different productivity monitoring techniques in building construction is very essential in helping researchers to know the different tools and techniques performed for data acquisition and analysis, limitations of each technique, the activities monitored by each technique and the possibility of integrated these techniques and tools to obtain highly efficient and accurate results. Furthermore, assessing the effective productivity monitoring tools and techniques in building projects is significant for construction managers to know the variability and uncertainty in project activities, which will help them in making effective decisions timely and accurately and that will lead to reducing costs, making timely corrective actions, reduce errors and project success.

2. Research methodology

A search of the literature was conducted using the systematic approach of the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) protocol guidelines and combined with bibliometric analysis. PRISMA produces evidence-based outcomes and, at the same time, enhances the reporting quality of the review with a transparent literature selection procedure (Moher *et al.*, 2009). Moreover, it has a methodological and analytic process that is clear and easy to understand (Moher *et al.*, 2015). Bibliometric methods allow the researchers to base their judgments on aggregated bibliographic data provided by other researchers who are working in the same field and have represented their viewpoints through collaboration, citation and writing. Also, bibliometric methods were complemented with meta-analysis and qualitative structured literature reviews, so a better reviewing and evaluating of scientific literature can be achieved (Zupic and Čater, 2015).

In this research, a systematic literature review approach was first applied to collect related information. This was done by isolating the collected papers according to their keywords, document types, language, etc. Second, the bibliometric analysis was conducted to improve the prominent research and identify the research trends using quantitative analysis.

2.1 Data acquisition

Based on the systematic review guidelines and the objectives of this study, the first phase was designed to gather the data in relation to the previous studies on construction productivity monitoring techniques for building construction projects. A protocol was established for collecting articles by overviewing the studies that have performed different techniques and tools for the productivity monitoring of construction building projects. The study scope was limited to monitoring the productivity of construction building projects publications, English publications and published between 2010 and 2020.

2.1.1 Databases and keywords. Four databases were selected to collect the available literature on the productivity monitoring techniques for construction building projects namely Web of Science (WoS), Scopus, Science direct and American Society of Civil Engineers (ASCE); all these databases were frequently adopted by researchers for gathering the literature (Al-Sabaeei *et al.*, 2020; Braun *et al.*, 2019; Hasan *et al.*, 2018; de Oliveira *et al.*, 2019; Ramirez Lopez and Grijalba Castro, 2021). Between these databases, WoS and Scopus databases are the largest citations and abstract database of the peer-reviewed literature that delivers an overview of the world's research output in the fields of technology, science, etc. (Braun *et al.*, 2019; de Oliveira *et al.*, 2019).

The keywords used in this study to identify the most relevant publications regarding the subject matter are: construction productivity, productivity monitoring, monitoring techniques and building projects. In WoS, the searching was performed by using advanced search with the designed keywords query as "TS=((productivity AND (monitoring OR measuring) AND construction) NOT (infrastructure or highway or tunnel))". The searching provided 976 results, which were refined to 843 results based on the study scope. For the Scopus database, the keywords inquiry was created by applying a combination as "TITLE-ABS-KEY((productiv*) AND ((monitor*) OR (measure*)) AND (building) AND ((construction) OR (project))) AND NOT ((infrastructure) OR (highway) OR (tunnel))". The searching provided 752 results, which were filtered down to 490 results based on the defined scope. In ScienceDirect, the keywords combination "TITLE-ABS-KEY (productivity) AND (monitoring) AND ((construction) OR (building)) AND NOT ((infrastructure) OR (highway) OR (tunnel))" was used and 170 publications were found, from which 126 publications were selected according to the defined scope. In the ASCE database, the search was performed by "TITLE (productivity AND (monitoring OR measuring) AND construction)" as keywords and 11 results were selected.

A total of 1,470 publications were collected under the defined protocol, which were published from 2010 to 2020. Figure 1 shows that researchers' interest in the issue of productivity monitoring is almost constantly increasing from 2010 to 2020. This confirms the significance of this topic in the construction sector. Also, the highest number of publications (215) was found in 2020, which indicates that the topic still very active and the general trend

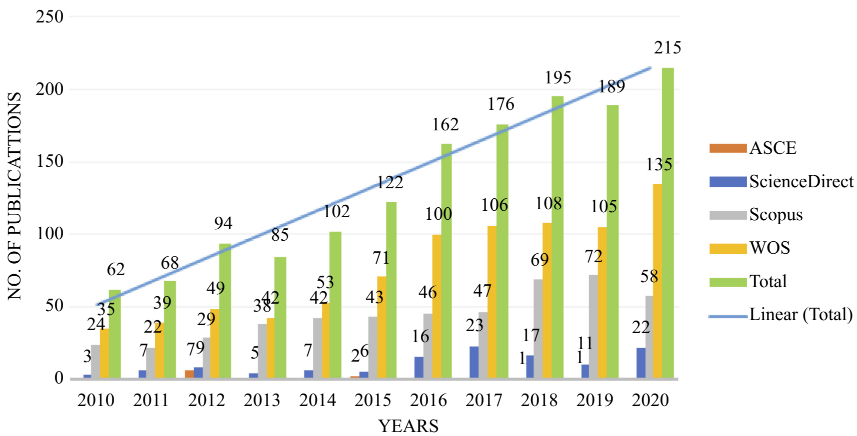

Figure 1.
Number of publications over the years

of the publication is increasing. The maximum data collection was obtained from the Web of Science (WoS) and Scopus databases.

2.2 Data filtration

This stage was divided into four steps, identification, screening, articles eligibility and included articles as shown in Figure 2. Where the first step is for identifying the number of publications from each database. The second step is the screening, where the publications were screened for duplication, title screening and abstract screening. The duplication was performed based on the articles with the same titles that got reselected again by other keyword combinations from the same database or the other databases, and 178 publications were found common or repeated between databases, which were eliminated. The irrelevant publications were excluded by performing title and abstract screening. The screening was conducted, where the irrelevant studies were excluded after reviewing each article for its title,

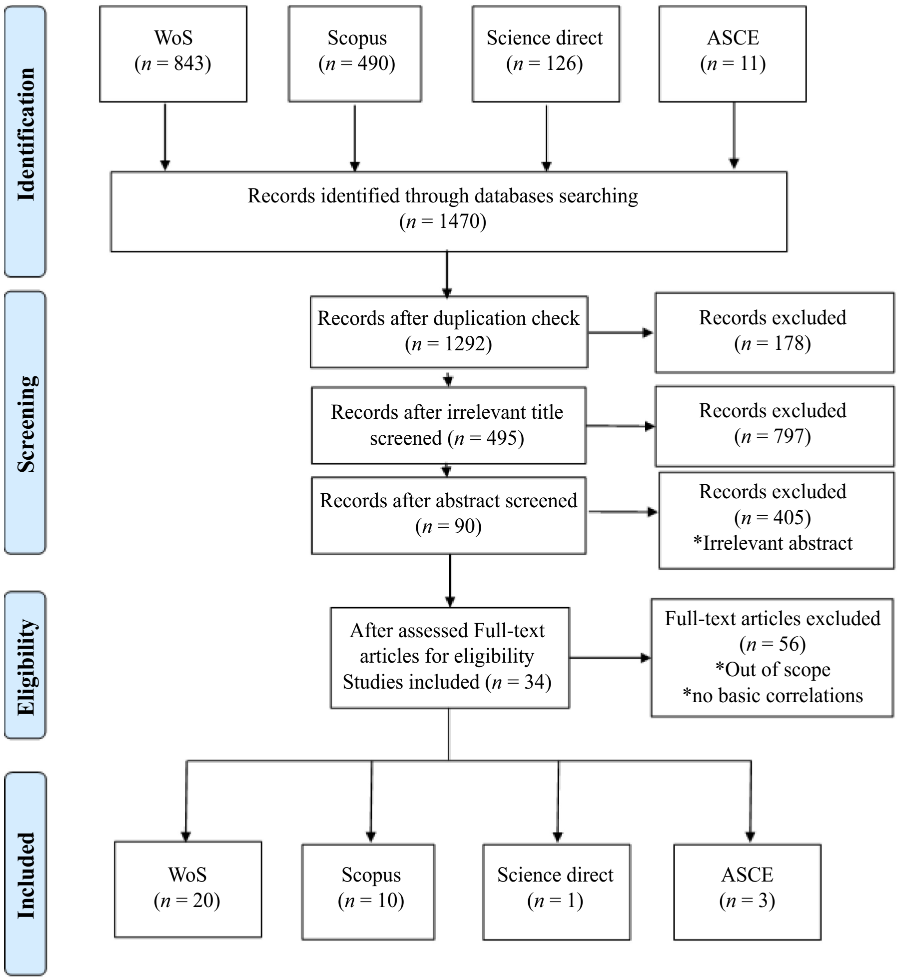

Figure 2.
PRISMA flowchart

where a total of 797 irrelevant publications were eliminated. Next, in further checking by reviewing the abstract of each article which led to eliminating 405 irrelevant publications.

While in the third step, the eligibility of full text for each article was explored thoroughly for its adopted tools and methodologies, which led to a total of 34 included publications, and 56 publications were excluded. Finally, the included publications number for each database after conducting the previously described filtering processes led to the following summary: 20 studies from WoS, 10 from Scopus, 3 from ACSE and 1 from ScienceDirect as shown in Figure 3.

2.3 Data processing and analyses

The bibliographic analysis examines the author's research areas, the paper content and the citation network among other things (Lim and Buntine, 2016). In this paper, the selected publications were examined according to the article source, author keywords and tools used for productivity monitoring. Finally, the map of the selected publications was examined based on the title and abstract to visualize the connections between terms and the most co-occurred terms. a strength, a clear overview and lack of author terms discussion can be provided by this visualization.

2.3.1 Publications distribution by year. From Figure 3, during the selected period from 2010 to 2020, it can be observed that the trend of publications increasing over time, where the maximum number of publications were found in 2017 and 2020 with eight and seven publications, respectively. One publication was found in 2010, 2012 and 2014, three in 2019, four in 2011 and 2015, and five publications were found in 2018. No publications were available in 2013 and 2016. Also, it is observed that WoS and Scopus are producing the maximum number of research articles related to construction productivity monitoring, and the interest of researchers in productivity monitoring is not lost, it is ascending during the time.

2.3.2 Publications distribution by journal/conference. Collected publications were also analyzed for their sources to find the distribution by journal/conference as shown in Figure 4. Among the 34 selected publications, just three publications were selected from three different conferences, while the rest 31 publications were found in journals. Also, it was found that "Journal of Construction Engineering and Management" has the highest number of publications among the selected journals/conferences with six records, then "Automation in Construction", "Journal of Asian Architecture and Building Engineering" and "Journal of Computing in Civil Engineering" with four, three and two records, respectively. While the rest of the journals/conferences recorded only one result.

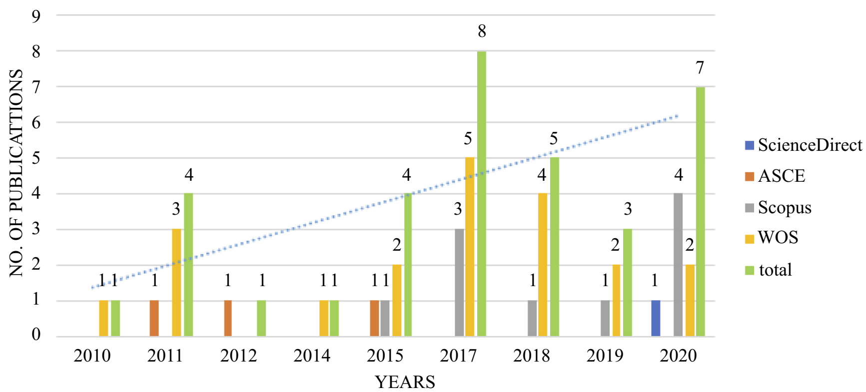

Figure 3.
Number of selected publications from databases

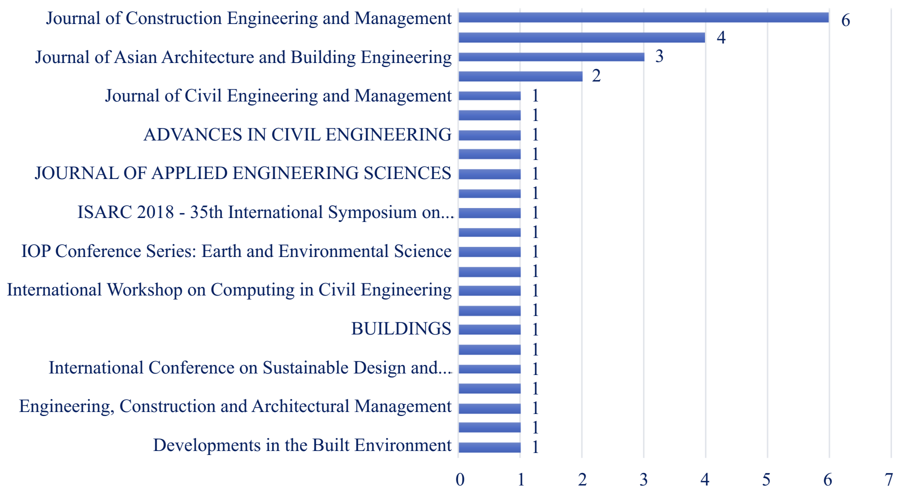

Figure 4.
Number of publications according to journal/conference distribution

2.3.3 Publications distribution by the building element. The building activities monitored in the selected data were analyzed as shown in Figure 5. It can be seen that the masonry works and formwork were the highest monitored activity among other activities; they have monitored alone or with other activities in eight articles, which represent about 23.5% of the selected studies. The second most popular activity among the selected results was the concrete work that was monitored alone or with other activities in six articles with about 17%. While the excavations operations and reinforcement activities were monitored in four articles with about 11.8%, and the worker movements were listed in three articles. Each of the plastering, curtain walls and construction machines activities were monitored in two articles only. The rest of the activities were monitored in just one article.

2.3.4 Publication distribution based on productivity monitoring methods. The methods used for monitoring productivity in construction were also analyzed. According to Figure 6, "manual observations and statistical methods" were the most popular methods as it was used 12 times in the results, which means that 35% of the selected studies have used this method,

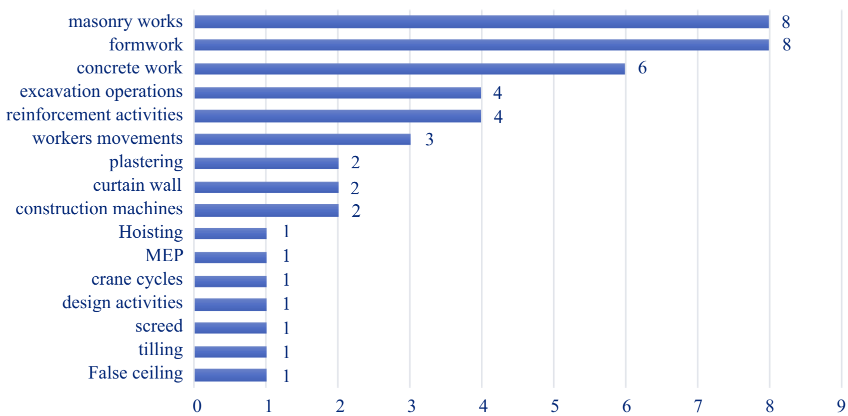

Figure 5.
Activities monitored distribution among the selected studies

the building information modeling (BIM) and vision-based methods came in the second place as each of them were repeated in eight times (23.5%) studies. While the artificial neural network (ANN) was repeated four times (11.7%) studies. Photogrammetry and support vector machine (SVM) methods were repeated two times (5.8%). Few studies were performed using the other methods (convolutional neural network, accelerometer-based method, smartphone sensors, GPS and fuzzy set theory, Bluetooth, simulation-based (WebCYCLONE) and Audio Signals) with 2.9%. It is worth noting that these methods were used either alone or in combination with other methods.

3. Results and discussion

3.1 Bibliometric mapping

With the recent growth of information technology, visualization techniques and scientific index, the bibliometric method provides researchers with a tool to visualize and understand the trends and the connections in the literature. The bibliometric examines invisible connections based on bibliographic records of the literature. A co-occurrence keyword analysis, for example, helps reveal methodologies and topics of the research in one research field (Su and Lee, 2010). Bibliometric mapping was performed for keywords analysis to visualize and understand the connections and trends between the selected studies.

3.1.1 Keywords analysis. Keywords analysis was conducted using VOSviewer software, which is a freely available software that was developed for viewing and constructing bibliometric maps. VOSviewer pays special interest to the graphical representation of bibliometric maps among other software that is used for bibliometric mapping (van Eck and Waltman, 2010). The primary contents of current studies and research topics in a specific field are represented by keywords. The co-occurrence of keywords shows the inter-closeness among them (Xia *et al.*, 2020). By setting the minimum occurrence of a keyword at two, in the selected 34 publications, 148 keywords were found, with 22 keywords meeting the threshold point. By reading the publications, some keywords that are related to the same meaning in language or logic were combined, such as “Building information modeling” and “BIM”, “artificial neural network” and “artificial neural networks” “ANN”, etc.

Figure 7 presents the final visualization of co-occurring keywords. The size of a node reflects a visual representation of the occurrence of a given keyword, and the thickness of the connecting lines indicates the interconnectedness between keywords. The occurrence of the keywords “productivity” and “construction” was the highest with 30 and 24 occurrences, respectively. Also, it can be seen that a large portion of studies studied the relationship between productivity and construction and that indicates the significance of productivity in the construction industry among studies.

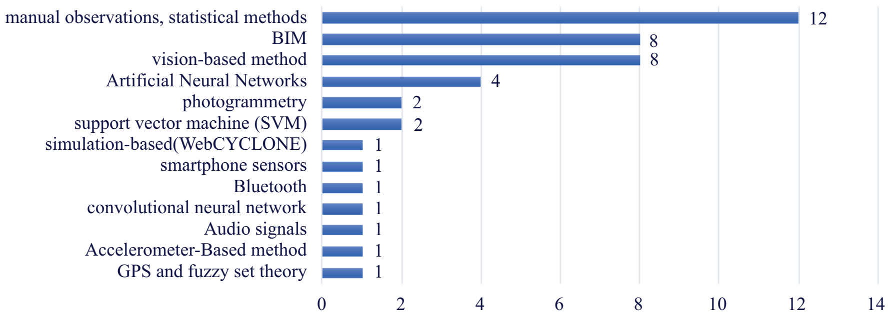

Figure 6.
Methods distribution among the selected studies

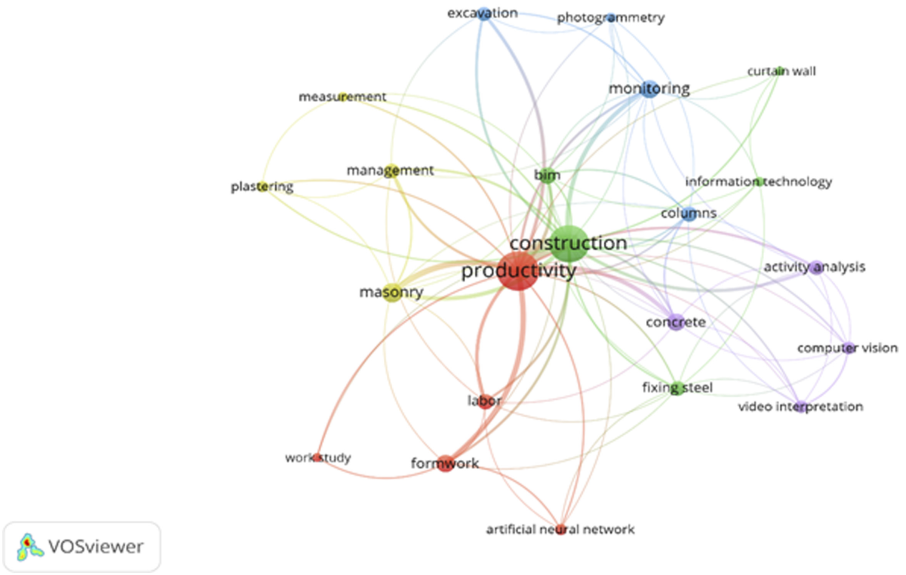

Figure 7. Co-occurrence mapping of keywords

Table 1 shows a summary of co-occurred keywords with the total link strength between the keywords. Among the different construction activities, masonry (Arif and Khan, 2020; Bhatti *et al.*, 2019; Chui *et al.*, 2012; Florez and Cortissoz, 2017; Joshua and Varghese, 2011; Mao *et al.*, 2018; Ouga *et al.*, 2020) and formwork (Azhar *et al.*, 2020; Bhatti *et al.*, 2019; El-Gohary *et al.*,

Table 1. Summary of occurrences and links between keywords

Keywords	Occurrences	Total link strength
Productivity	30	76
Construction	24	69
Masonry	8	20
Formwork	6	17
Monitoring	5	18
Concrete	5	17
BIM	5	16
Labor	4	13
Activity analysis	4	12
Excavation	4	11
Columns	3	13
Fixing steel	3	12
Management	3	12
Computer vision	2	9
Video interpretation	2	9
Artificial neural networks	2	7
Plastering	2	7
Information technology	2	6
Measurement	2	6
Photogrammetry	2	6
Curtain wall	2	5
Work study	2	5

2017; Golnaraghi *et al.*, 2019; Lee *et al.*, 2017; Mine *et al.*, 2015) were presented lonely or combined with other activities, the most activities monitored for their productivity.

3.2 Productivity monitoring tools

Productivity in the construction sector is known as an indicator of the consistency and efficiency of construction firms. There have been numerous studies on the effective analysis, measurement and monitoring of productivity data, which resulted from the growing interest in building productivity (Lee *et al.*, 2014). Monitoring and measuring construction productivity are crucial for project success. Also, they can improve the sustainability and competitive advantages of construction firms in the market by adopting the best standard productivity practices (Marzuki *et al.*, 2012). Most of the factors that affect construction productivity can be improved by monitoring productivity effectively (Konstantinou and Brilakis, 2016). According to the selected studies, the monitoring methods can be divided into two major methods: traditional methods and automated methods. The traditional method depends on manual observations and statistical methods for collecting and analyzing the productivity data (Kim and Kim, 2018; Vahdatikhaki *et al.*, 2015). While automated methods depends on a new technologies and artificial intelligence for monitoring different aspects of construction operations (Sherafat *et al.*, 2020). Table 2 shows the studies, activities monitored and the methods used in these studies.

Although traditional methods have some limitations for site monitoring, such as time-consuming, operose, missing or incomplete information, resulting in a slow flow of information, make effective decision-making difficult, labor-intensive and high expenses (Kimoto *et al.*, 2005; Luo *et al.*, 2018; Naoum, 2015; Omar *et al.*, 2018; Yang *et al.*, 2015), they have been used in several studies to monitor the productivity of different indoor activities (such as tiling, plastering, masonry work, false ceiling boarding and fixing and application of screed) and outdoor activities (such as formwork, reinforcement steel activities, MEP, concrete work and curtain wall) (Alchaer and Issa, 2020; Azhar *et al.*, 2020; Bhatti *et al.*, 2019; Chui *et al.*, 2012; Florez and Cortissoz, 2017; Gouett *et al.*, 2011; Hwang, 2010; Ko and Han, 2015; Kubečková and Smugala, 2020; Mao *et al.*, 2018; Mine *et al.*, 2015; Ouga *et al.*, 2020).

To overcome the limitations in traditional monitoring methods, several studies have made valuable attempts by adopting modern technologies and artificial intelligence to monitor, measure and analyze construction productivity (Arif and Khan, 2020; Golnaraghi *et al.*, 2019; Konstantinou *et al.*, 2019; Petrov and Hakimov, 2019). Nowadays, modern digital technologies used in construction monitoring are supporting the user to resolve issues like; time-consuming, slow flow of information, labor-intensive and high expenses of collecting data (Petrov and Hakimov, 2019). Also, the advent of high storage databases, high-resolution cameras and accessibility of the internet has transformed the ongoing construction operations' documentation methods into a paradigm shift. As a result, studies in the field of computer vision-based techniques have grown increasingly and become popular among researchers to monitor construction site activities (Khosrowpour *et al.*, 2014; Luo *et al.*, 2018; Zhu *et al.*, 2016).

Computer vision is the automated extraction of information from images. Information can mean anything from 3D models, camera position, object detection and recognition to grouping and searching image content (Solem, 2012). Vision-based techniques combined with BIM are used to monitor the productivity of construction activities (such as formwork, crane cycles, excavation operations, concrete work, earthmoving, Scaffold installation, slab pour, column pour and hoisting activities) (Forsythe and Sepasgozar, 2019; Lee *et al.*, 2014; Liu and Golparvar-Fard, 2015; Ren *et al.*, 2017). Although these techniques have achieved many improvements in monitoring construction productivity compared to traditional methods, the challenging environment of construction sites affects the accuracy of these techniques. The accuracy of monitoring construction productivity methods is affected by congestion,

Table 2. Summary of methods used for monitoring construction productivity

Title	Authors	Method	Activities	Remarks
Cross-validation of short-term productivity forecasting methodologies	Hwang (2010)	Manual observations and statistical methods	Concrete work	Time-consuming, not applicable for long time productivity estimation, labor intensive, not applicable for ongoing operations and slow information flow
An object recognition, tracking and contextual reasoning-based video interpretation method for rapid productivity analysis of construction operations	Gong and Caldas (2011)	Vision-based method	Earthmoving, scaffolding, slab, column pouring and hoisting	Good in tracking single object, occlusion problem and limited in view
Activity analysis for direct-work rate improvement in construction	Gouett *et al.* (2011)	Manual observations and statistical methods	Workers activities	Time-consuming, labor-intensive and slow information flow
A visual monitoring framework for integrated productivity and carbon footprint control of construction operations	Heydarian and Golparvar-Fard (2011)	Vision-based and BIM	Excavation operations	Limited in view due to the fixed camera
Accelerometer-based activity recognition in construction	Joshua and Varghese (2011)	Accelerometer-based method	Masonry work	Accuracy of 80% was obtained, attaching wired accelerometers to the waist of a mason limit the movement of the worker due to the weird sensor attached to the laptop
On-site labor productivity measurement for sustainable construction	Chui *et al.* (2012)	Manual observations and statistical methods	MEP and masonry work	Time-consuming, labor intensive and slow information flow
A system model for analyzing and accumulating construction work crew's productivity data using image processing technologies	Lee *et al.* (2014)	Vision-based method and BIM	Formwork	Limited to the camera view
Applying artificial neural networks for measuring and predicting construction-labor productivity	Heravi and Eslamdoost (2015)	Artificial neural networks and manual observation	Concrete work	Does not take all predicting factors to develop the model which could affect the accuracy of the model
Development of construction performance monitoring methodology using the bayesian probabilistic approach	Ko and Han (2015)	Manual observations and statistical methods	Curtain wall	Time-consuming, labor intensive, not applicable for a large construction site, and slow information flow

(*continued*)

Table 2.

Title	Authors	Method	Activities	Remarks
Crowdsourcing construction activity analysis from jobsite video streams	Liu and Golparvar-Fard (2015)	Vision-based method	Concrete work	Obtained 85% accuracy, nonexpert annotators from a crowdsourcing marketplace negatively affect the quality of the assessment results
An observational study on the productivity of formwork in building construction	Mine *et al.* (2015)	Manual observations and statistical methods	Formwork	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
Fusion of photogrammetry and video analysis for productivity assessment of earthwork processes	Bügler *et al.* (2017)	Photogrammetry	Excavation operations	Obtained 90% accuracy, poor visual conditions created by the occlusion and bad weather led to higher errors
Activity analysis of construction equipment using audio signals and support vector machines	Cheng *et al.* (2017)	Audio signals, support vector machine (SVM)	Construction machines	Applicable only for construction machines that generate discrete sound patterns during their routine operations and analysis activities of single machines, make it unproper in crowded sites
Engineering approach using ANN to improve and predict construction labor productivity under different influences	El-Gohary *et al.* (2017)	Artificial Neural Networks, manual observation	Formwork, fixing steel	Obtained high accuracy, depending on factors of productivity to develop the prediction model affect the model due to unstudied factors
Using workers compatibility to predict labor productivity through cluster analysis	Florez and Cortissoz (2017)	Manual observations and statistical methods	Masonry work	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
Framework for the validation of simulation-based productivity analysis: focused on curtain wall construction process	Han *et al.* (2017)	Simulation-based (WebCYCLONE), manual observation	Curtain wall	Time-consuming, not providing real-time monitoring and the slow flow of information
BIM-assisted labor productivity measurement method for structural formwork	Lee *et al.* (2017)	BIM and manual observations	Formwork	Time-consuming
Construction productivity and ergonomic assessment using mobile sensors and machine learning	Nath *et al.* (2017)	Smartphone sensors and support vector machine (SVM)	Workers movements	Obtained 80% accuracy, costly due to the required wearable sensors attached to each worker

(*continued*)

Title	Authors	Method	Activities	Remarks
Automated monitoring of the utilization rate of onsite construction equipment	Ren *et al.* (2017)	Vision-based method	Excavation equipment	Fixed camera, limited in view, and occlusion affect the quality of monitoring
Automation of measuring actual productivity of earthwork in urban area, a case study from montreal	Alshibani (2018)	GPS	Excavation work	It also assumes that the tracked fleet consists of trucks of the same size, which may not always be true, GPS receivers should be mounted on all trucks, which makes the proposed methodology expensive
Convolutional neural networks: computer vision-based workforce activity assessment in construction	Luo *et al.* (2018)	The convolutional neural network, vision-based	Reinforcement activities	Achieves an average accuracy of 85%, attention needs to focus on addressing occlusions
Using wearable devices to explore the relationship among the work productivity, psychological state and physical status of construction workers	Mao *et al.* (2018)	Manual observations, statistical methods	Masonry work	Time-consuming, labor intensive, not applicable for a large construction site, slow information flow and costly due to the required wearable sensors attached to each worker
Toward an automated photogrammetry-based approach for monitoring and controlling construction site activities	Omar *et al.* (2018)	Photogrammetry and BIM	Columns	Obtained high accuracy, required a lot of fixed cameras for monitoring (12 cameras)
BIM log mining: measuring design productivity	Zhang *et al.* (2018)	BIM and manual survey	Design productivity	Applicable only for comparing the production rate between designers within the same level of quality and design excellence
Building construction labor productivity in arid climate environment	Bhatti *et al.* (2019)	Manual observations, statistical methods	Masonry, fixing steel and formwork	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
Measuring installation productivity in prefabricated timber construction	Forsythe and Sepasgozar (2019)	Vision-based method	Crane cycles	Fixed cameras, time consuming, manually measures and analyses the productivity from the registered video
Application of artificial neural network(s) in predicting formwork labor productivity	Golnaraghi *et al.* (2019)	Artificial neural networks and manual observation	Formwork	Does not take all predicting factors to develop the model, which could affect the accuracy of the model

(*continued*)

Table 2.

Title	Authors	Method	Activities	Remarks
Engineering productivity measurement: a novel approach	Alchaer and Issa (2020)	Manual observations and statistical methods	Screed, tiling, plastering and false ceiling activities	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
A real-time productivity tracking framework using survey-cloud-bim integration	Arif and Khan (2020)	BIM and manual survey	Concreting columns, concreting slab and masonry work	The accuracy depends on the quality of surveying
Strategy of change construction method to increase productivity and reduce waste in the private university buildings	Azhar *et al.* (2020)	Manual observations and statistical methods	Formwork	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
A study on 3D/BIM-based on-site performance measurement system for building construction	Cha and Kim (2020)	BIM and manual observation	Steel erection work	The data-capturing process of the proposed system remains manual-based
Statistical methods applied to construction process management	Kubečková and Smugala (2020)	Manual observations and statistical methods	Plastering and masonry work	Time-consuming, labor intensive, not applicable for a large construction site and slow information flow
Identifying productive working patterns at construction sites using BLE sensor networks	Mohanty *et al.* (2020)	Artificial neural networks and Bluetooth low energy	Movement and location of workers	Accuracy tracking in the range of 5–10 m (limited in range), tagged devices arises a personal private issue and idle time could not be recognized
Modelling block laying productivity on building sites in Kampala	Ouga *et al.* (2020)	Manual observations and statistical methods	Masonry work	Time-consuming, labor intensive, not applicable for a large construction site and slow feedback for management

Table 2.

background clutter, occlusions and illumination variations in construction sites (Konstantinou *et al.*, 2019; Luo *et al.*, 2018).

Photogrammetry is a technique that combines photogrammetry and computer vision algorithms to extract 3D geometries from a series of digital 2D images (Dering *et al.*, 2019). Photogrammetry integrated with BIM can be used to monitor reinforced concrete column pouring (Omar *et al.*, 2018) and excavation operations productivity (Bügler *et al.*, 2017). Although this technique can achieve a higher level of accuracy and considered as most economical among other monitoring techniques (Mahami *et al.*, 2019; Park *et al.*, 2019), it is affected by bad weather and poor visual conditions created by the occlusion led to higher errors (Bügler *et al.*, 2017; Omar *et al.*, 2018).

Tools such as audio signals, bluetooth and smartphone sensors combining with machine learning algorithms like ANN and SVM have been adopted by several studies to monitor the productivity of construction activities like (formwork, fixing steel, concrete work, movement and location of workers, machines activities and excavation operations) (Cheng *et al.*, 2017; El-Gohary *et al.*, 2017; Golnaraghi *et al.*, 2019; Heravi and Eslamdoost, 2015; Heydarian and Golparvar-Fard, 2011; Mohanty *et al.*, 2020, 2020; Nath *et al.*, 2017). Although adopting these tools with machine learning algorithms has enhanced their abilities to monitor construction productivity and achieved a good performance, attention is still needed to focus on addressing occlusions, the limited range for monitoring and personal privacy issues due to tagged devices. The provided information should be detailed not just "productive" or "unproductive" to help managers to make good decisions.

The accelerometer-based method was used to monitor masonry activities and an accuracy of 80% was obtained (Joshua and Varghese, 2011). Attaching wired accelerometers to the worker will result in a limited movement of the worker due to the weird sensors which should be attached to a laptop. Also, GPS has been used to monitor excavation activities and a good accuracy level was obtained (Alshibani, 2018), but it is limited to monitoring one excavator at a time. It is also assumed that the tracked trucks have the same size, which may not always be correct. In such cases, more GPS receivers should be mounted on all trucks on the site, which makes the proposed methodology expensive. BIM is a digital tool known as a platform for central integrated design, modeling, asset planning running and cooperation in the construction industry (Poljansek, 2018). Recently, there has been increasing attention for using BIM to stimulate multifaceted transitions in aspects of construction management (Borrmann *et al.*, 2018). As researchers aim to accomplish a higher level in terms of automation in construction, the applications of BIM have increased, almost BIM has been integrated with all the above-mentioned methods (computer vision-based, machine learning, photogrammetry and manual observations) (Arif and Khan, 2020; Cha and Kim, 2020; Heydarian and Golparvar-Fard, 2011; Lee *et al.*, 2014, 2017; Omar *et al.*, 2018; Zhang *et al.*, 2018) to monitor the productivity of different construction activities.

For the effective application of each of these methods on the job site, five key performance measures should be satisfied (Sherafat *et al.*, 2020): (1) accuracy, (2) computational time, (3) required devices, (4) appropriate special, temporal and weather conditions and (5) the range of activities it can cover. Also, nonintrusive for the workers (privacy issues) and the applicability of indoor-outdoor should be taken into consideration (Konstantinou *et al.*, 2019).

Table 3 provides a comparison of methods based on the selected studies. It can be observed that traditional monitoring methods have covered all the activities of construction sites. However, these methods have several limitations, such as time-consuming, the time and effort required for data collecting, skilled workers required to monitor and analyze productivity (labor-intensive), high expense and the slow flow of information which makes it difficult to take effective decisions. Computer vision has

Table 3. Comparison of implemented construction productivity monitoring methods

Method/ metrics	Range covered	Weather conditions	Computational time	Required devices, special requirements	Intrusiveness	Accuracy	Testing environment	Monitored activities
Traditional methods	Wide range	Affected	Slow	Skilled workers doing the survey	Non	High	Indoor-outdoor	Formwork, Steelwork, concrete work, masonry, screed, tiling, plastering, false ceiling, curtain wall, MEP and worker activities
Computer vision-based	Within cameras view	Affected	Fast	Cameras	Non	High- above 90%	Outdoor	Excavation, crane cycles, concrete work, hoisting, formwork and excavation equipment
Photogrammetry	Within cameras view	Affected	Fast	Cameras	Non	High- above 90%	Outdoor	Excavation operations and reinforced column
Bluetooth low energy	range of 5–10 m	Not affected	Fast	BLE tags, receivers, WIFI router and monitoring device (laptop)	Intrusive	High within the covered range	Lap	Movement and location of workers
Smartphone sensors	Narrow	Not affected	Fast	Smartphone sensors	Intrusive	80%	Lab	Worker movements
GPS	Wide	Not affected	Fast	GPS receivers	Intrusive	High	Outdoor	Excavation
Audio-based	One machine	Not affected	Fast	Microphone array	Non	80%–85%	Outdoor	Construction machines
Accelerometer-based	Narrow	Not affected	Fast	Laptop mounted with camera, accelerometer sensor	Intrusive	80%	Lap	Masonry work

covered a wide range of activities, high accuracy, low cost and fast in extracting data. Computer vision depends heavily on fixed cameras mounted in the construction site which makes it limited to the view range of these cameras and the occlusion resulted from moving objects and workers on the site. Photogrammetry also has achieved high accuracy, low cost and fast information collecting and analyzing, but it is limited in view range due to the usage of fixed cameras.

Accelerometer-based, smartphone sensors and Bluetooth are low energy methods that can cover a small range distance for monitoring. The high initial cost due to the required devices and the tags mounted to the worker body make these methods intrusive, uncomfortable to the worker and not applicable in large-scale projects. Although GPS can cover a wide outdoor range distance for monitoring, its high cost due to the required devices, tags mounted to the worker body make these methods intrusive and uncomfortable for the worker, and not applicable in large-scale projects and indoor activities. Audio-based methods are applicable only for construction machines that generate discrete sound patterns during their routine operations and can analyze activities of a single machine, which makes it improper in crowded sites.

To overcome the limitations of each method, some researchers have integrated more than one tool or technology to get a more effective monitoring method. Table 4 shows some of the tools and technologies that have been integrated together and the monitored activities. For instants, BIM is integrated with computer vision-based (Lee *et al.*, 2014), photogrammetry (Omar *et al.*, 2018), manual observations (Arif and Khan, 2020; Cha and Kim, 2020; Lee *et al.*, 2017; Zhang *et al.*, 2018) to monitor construction productivity for formwork, excavation, columns, slab pouring, steel fixing and design activities. ANN has been integrated with manual observations (Golnaraghi *et al.*, 2019) to monitor the productivity for formwork, steel, concrete pouring activities and also integrated with low energy bluetooth to monitor the location of workers (Mohanty *et al.*, 2020). SVM has been integrated with audio-based (Cheng *et al.*, 2017) to monitor the construction machines that make noises during their activities and also integrated with smartphone sensors (Nath *et al.*, 2017) to predict the productivity of workers based on their ergonomic status. However, despite the values added by these studies, most of them still at the proof-of-concept stage. These studies did not cover all complex construction job sites and applied based on a small sample of construction workers and machines separately. Also, the number of implemented tools for monitoring (e.g. tags or cameras) was limited. Studies that cover the whole construction site with multiple machines and workers working concurrently will serve to better understand the significance of these studies in practice.

3.3 Conceptual framework

Based on the review of the selected studies, a conceptual framework was developed and presented in Figure 8. This framework shows the connectivity of productivity monitoring with productivity improvement, effective decisions and project success.

The conceptual framework illustrates that the use of new technologies and tools for monitoring productivity will result in an effective monitoring way. This will lead to collect the data about the construction site accurately, effectively and in real-time which will help the management to take effective decisions. Making effective decisions will lead to reduce costs, timely corrective actions and reduce errors. All these items surely will increase the chance of having successful projects. Having successful projects will enhance the firm's competitiveness advantages in the markets, develop the economy and achieve customer satisfaction. Therefore, much attention is required to monitor construction productivity effectively and using new techniques, which is linked directly to productivity improvements and construction project success.

Table 4.
The combined tools/technologies and activities monitored

Tools/ Technologies	BIM	CV	Photogrammetry	Traditional methods	ANN	SVM	Audio-based	Bluetooth	Smartphone sensors
BIM	–	Formwork and excavation	Columns	Columns, slab, steel, design and masonry	–	–	–	–	–
CV	Formwork and excavation	–	–	–	–	–	–	–	–
Photogrammetry	Columns	–		–	–	–	–	–	–
Traditional methods	Columns, slab, steel, design and masonry	–	–		Formwork, steel and concrete pouring	–	–	–	–
ANN	–	–	–	Formwork, steel and concrete pouring	–	–	–	Workers' location	–
SVM	–	–	–	–	–		Machines	–	Workers' movement
Audio-based	–	–	–	–	–	Machines	–	–	–
Bluetooth	–	–	–	–	Workers' location	–	–	–	–
Smartphones Sensors	–	–	–	–	–	Workers' movement	–	–	–

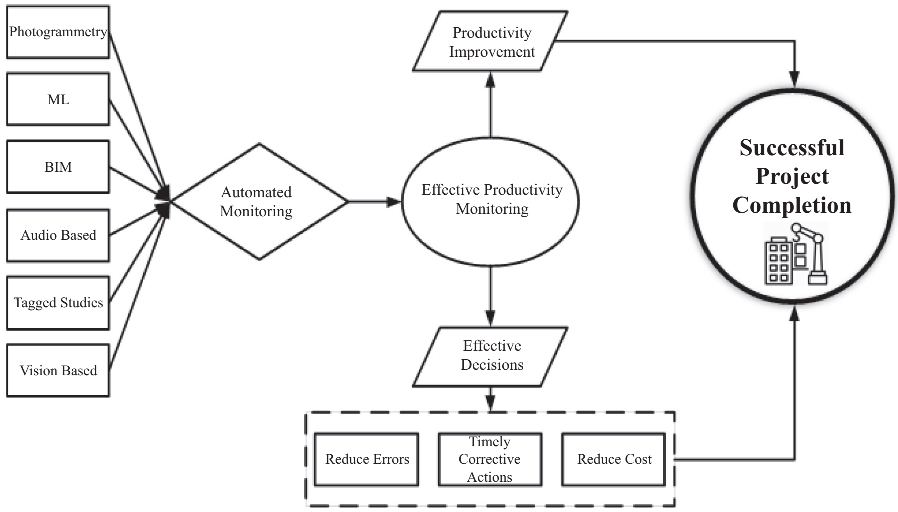

Figure 8.
Conceptual framework

4. Conclusion

Productivity in the construction sector is used as a pointer to the consistency and efficiency of construction firms. Selecting the most effective monitoring and measuring of construction productivity methods is crucial for project success. The monitoring methods can be divided into two major methods, traditional methods and automated methods. Although traditional methods have several limitations, they were the most popular among the selected studies. The second and the third popular in automated methods were computer vision-based and photogrammetry; these techniques provide promising results with high accuracy, fast information flow and low cost. But they depend heavily on fixed cameras that make the range of view to be limited. Also, tagged technologies like (GPS, smartphone sensors, bluetooth and accelerometer-based) were used to monitor the productivity of equipment and workers' activities. The main limitations of tagged technologies were the limited cover range and intrusive issues due to the mounted tags on the worker body. Although audio-based technologies were used to monitor productivity, it is not applicable in crowded construction sites and not suitable for monitoring tools that do not make a noise while performing various tasks like cranes. Machine learning algorithms (ANN, SVM) and BIM were integrated with monitoring tools and technologies by researchers to enhance the monitoring performance and to increase the automation in monitoring construction productivity. Although the researchers made an effort in this field, it was observed that the current studied techniques did not cover all complex construction job site and were applied based on a small sample of construction workers, operations and machines separately. Also, the number of implemented tools for monitoring (e.g. tags or cameras) was limited. Future research may focus on monitoring the whole construction site with multiple workers and machines working concurrently.

This study can be considered to have certain limitations. This study covers only productivity monitoring on building construction projects during the period from 2010 to 2020. However, the methods that have been used for monitoring productivity in other construction projects like dams, tunnels and highway infrastructure were not investigated. Besides, data were collected only from four different databases, i.e. WoS, Scopus, ScienceDirect and ASCE. Therefore, due to the variation of information in the exported data from these databases, a few bibliometric analyses were not performed such as author co-citation analysis, journal analysis, institutional analysis, countries analysis, etc.

5. Future research direction

Although researchers made significant efforts for productivity monitoring, several challenges are still to overcome for maximum extraction of information, and a few are being highlighted in this study:

(1) The major current practices for productivity monitoring are depending on traditional monitoring techniques which have a lot of limitations. It is beneficial to make more effort to adopt automated monitoring technologies to overcome the limitations of traditional methods.

(2) Few studies made significant efforts in adopting automated productivity monitoring, but it was observed that the data-capturing process of the used systems remains manual-based. The construction job site is unsteady and changeful, and data measurement is time- and effort-consuming. Thus, a new technology for data capture and measurement will be very beneficial. for instance, considering drone technologies and image analyses, the systems would become more powerful and efficient (Cha and Kim, 2020).

(3) The systems used for monitoring need more development to be implemented in the actual construction sites. Investigating the several limitations and factors that affect the quality, accuracy and effectiveness of each technique used for productivity monitoring will be necessary to enlarge the efficiency of monitoring in the construction sites.

(4) Although the indoor activities represent a high-ranking part of any construction project, the average share of indoor activities in the total budget lies between 25% and 40%, the available techniques for monitoring mostly deal with outdoor construction activities (Kropp *et al.*, 2018). Therefore, the study of productivity monitoring of indoor activities like masonry, plastering, painting, doors and windows will be very valuable to enable the project managers to make effective decisions, which will result in cost savings and avoiding time overruns in construction projects.

(5) The financial impact is one of the barriers to the successful implementation of traditional and automated monitoring techniques. So, for the significant performance of each technique, less expensive methodologies are to be promoted in the construction sites.

(6) Each technique has its advantages and limitations. It is beneficial to integrate possible techniques to overcome the deficiencies of each one for maximum extraction of information.

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